Sudhanvi Vakul Vakulabharanam

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SUMMARY

RTL / Digital Design Engineer with 2+ years of hands-on experience in SystemVerilog RTL development, FPGA implementation, and SoC-oriented hardware design, backed by over 5 years of broader engineering experience across industry, research, and applied technical environments. Skilled in block-level microarchitecture, AXI/APB integration, clock/reset design, RTL simulation, waveform debug, linting, synthesis, static timing analysis, and timing closure. Hands-on experience building FPGA-based accelerators and embedded hardware systems, with working knowledge of UVM-based verification, structured testbench development, hardware-software debugging, and ASIC-oriented design methodologies.

SKILLS

Hardware Description Languages: SystemVerilog, Verilog, VHDL; **Programming Languages:** Python, C/C++, TCL, Embedded C
RTL Design: Block-Level Microarchitecture, FSM Design, AXI/APB Integration, Clock/Reset Design, CDC, SoC Integration, DMA, Pipelined Datapaths, Fixed-Point Design, Power/Area Trade-offs
Verification: RTL Simulation, Self-Checking Testbenches, SVA, Waveform Debug, Linting, UVM Methodology, cocotb
Protocols: AXI, APB, UART, SPI, I2C, Ethernet; **Platforms:** Basys3, Kintex-7 KC705, PYNQ-Z2
Tools: Xilinx Vivado, ModelSim, Verilator, OpenSTA, GTKWave, Git, Yosys, Synopsys Design Compiler, VCS

WORK EXPERIENCE

ELECTRONICS DESIGN ENGINEER (Project Manager) – Full-Time | Community Dreams Foundation | September 2025 – Present

- Founded and led VLSIVault, a structured hardware engineering platform focused on RTL design, microarchitecture, digital verification, and ASIC/FPGA-oriented system development.
- Architected a multi-track curriculum spanning RTL microarchitecture, processor design, communication interfaces, low-power architecture, functional verification, DFT concepts, and physical design workflows.
- Designed synthesizable SystemVerilog RTL for datapath and control logic with emphasis on FSM design, clock/reset robustness, CDC-safe design, and interface integration.
- Developed verification environments using self-checking testbenches, SVA, cocotb-based components, functional coverage, and regression-style simulation.
- Supported RTL simulation, synthesis, and static timing analysis while building reusable simulation, debug, documentation, and project infrastructure for FPGA and ASIC-oriented workflows.

RESEARCH ASSISTANT – Part-Time | Illinois Institute of Technology, Chicago | September 2024 – February 2025

- Evaluated edge-AI architectures including MobileViT, EfficientFormer, and Swin-T, benchmarking accuracy and latency across heterogeneous compute targets including GPU, TPU, and embedded SoC platforms.
- Implemented model quantization and graph optimization to enable efficient deployment on embedded hardware platforms, improving inference efficiency and resource utilization.
- Developed an FPGA-based inference workflow on the PYNQ-Z2 platform, implementing weight formatting, test-vector generation, and hardware/software integration for inference execution.
- Debugged hardware-software mismatches using waveform tracing, intermediate tensor validation, and pipeline inspection during FPGA inference deployment.

SYSTEM ENGINEER – Full-Time | Tata Consultancy Services | Hyderabad, India | August 2020 – July 2023

- Developed Python and SQL-based automation scripts and reporting workflows to improve validation efficiency, streamline production support, and reduce manual effort.
- Built operational dashboards and process automation solutions in collaboration with QA and cross-functional teams to improve reliability, issue tracking, and workflow visibility.

PROJECT APPRENTICE – Full-Time | DRDO | Hyderabad, India | January 2019 – April 2020

- Designed a two-way UART-to-Ethernet module in Verilog RTL using Vivado IP Integrator and integrated it into a PCIe-based data path.
- Implemented UART packetization, buffering, and clock-domain-safe data handoff to support reliable UART-to-Ethernet bridging under burst-traffic conditions.
- Verified functionality through simulation and FPGA board validation using TeraTerm for UART and Wireshark for Ethernet; applied SDC constraints, STA, and timing closure across multiple baud rates and traffic conditions and performed KC705 bring-up via JTAG with waveform/log-based corner-case debug.

PROJECT APPRENTICE – Full-Time | Bharat Dynamics Limited | Hyderabad, India | April 2018 – June 2018

- Built a MATLAB-based RF/Radar analysis tool for through-wall detection, modeling attenuation, propagation loss, multipath/scattering, and Doppler effects to evaluate signal behavior in obstructed environments.
- Improved detection performance by analyzing receiver SNR, clutter suppression techniques, and threshold-parameter sweeps to study trade-offs in sensitivity and false detections.

PROJECT EXPERIENCE

RTL Design Portfolio: CDC, Mini-SoC Integration, and Low-Power Architecture

SystemVerilog | AXI/APB | Vivado/XSIM/ModelSim | Verilator | OpenSTA | GTKWave | Python | Git

- Developed a staged RTL design portfolio covering repeatable simulation/debug workflows, CDC-safe and reset-aware RTL, mini-SoC subsystem integration, and low-power digital architecture.

- Designed synthesizable SystemVerilog RTL for FSM-based control logic, datapath modules, memory-mapped registers, and multi-clock subsystems, with emphasis on clock/reset architecture, synchronizer-based CDC, and microarchitectural partitioning.
- Implemented SoC-style digital subsystems using AXI/APB control paths, MMIO register interfaces, timer/interrupt logic, and protocol-connected modules for structured subsystem integration.
- Built reusable verification and implementation analysis flows using self-checking testbenches, SVA, cocotb, waveform-driven debug, lint-style RTL checks, synthesis, and STA to support reproducible RTL development.

FPGA-based Systolic Array Hardware Accelerator Design of VGG11

C++ | Vitis HLS | AXI4 | AXI-Lite | FPGA Acceleration | HLS C-Simulation

- Architected a systolic-array processing-element microarchitecture for VGG11 in C++ using Vitis HLS, applying loop pipelining, unrolling, array partitioning, tiling, and kernel grouping to improve parallelism, throughput, and data reuse.
- Implemented AXI4 memory-mapped data interfaces for feature maps, weights, bias, and output buffers with AXI-Lite control, enabling structured accelerator integration and hardware-oriented data movement.
- Validated functional correctness through HLS C-simulation against golden vectors and performed design-space exploration across PE count, tile size, and kernel grouping to optimize latency and architectural efficiency.
- Analyzed HLS synthesis QoR reports for latency, DSP, BRAM, LUT, and FF utilization, evaluating performance-resource trade-offs relevant to FPGA-based accelerator microarchitecture.

Low-Power 10T SRAM Using Sleep Transistors and Drain Gating Techniques for Improved Performance

Cadence Virtuoso | GPDK45 | Spectre | HSPICE | SRAM Design | Low-Power VLSI

- Designed a 10T SRAM bitcell in Cadence Virtuoso (GPDK45) using Sleepy Keeper, MTCMOS, and drain-gating techniques to reduce leakage, standby, and dynamic power while maintaining read, write, and hold stability.
- Ran Spectre and HSPICE transient, DC, and leakage simulations to characterize read/write delay, retention leakage, and static noise margin across PVT conditions.
- Benchmarked the proposed cell against conventional 6T and 10T SRAM cells, quantifying power, delay, and stability trade-offs across voltage and temperature corners.

Implementation of 6T SRAM Cell using ASAP7 7nm Technology in Cadence Virtuoso

Cadence Virtuoso | ASAP7 | HSPICE | FinFET | SRAM Design

- Implemented and simulated a 6T SRAM bitcell in ASAP7 7nm FinFET technology for read, write, and hold operation using HSPICE and waveform-based functionality debug.
- Analyzed low-power SRAM behavior with emphasis on gate leakage, subthreshold leakage, and read/write path trade-offs in scaled technology nodes.
- Compared baseline and optimized cell behavior, measuring dynamic power and documenting observations related to low-power memory design.

Other Projects

- Implemented DAB/DN/DINO-DETR in PyTorch; benchmarked mAP/AR on PASCAL-VOC and profiled inference accuracy–latency trade-offs.
- Built a 4WD Arduino robot (Embedded C) with UART Bluetooth, PWM motor control (L298N), and sensor-based obstacle avoidance, validated latency and control stability.
- Built YOLOv2 training and inference pipeline in Python (GPU); analyzed accuracy vs compute latency trade-offs.

EDUCATION

Illinois Institute of Technology, Chicago, IL, USA. August 2023 – May 2025

Master of Science | Electrical and Computer Engineering

Relevant Coursework: RFIC, VLSI Design, CMOS Circuits, Computer Architecture & Design, Hardware Acceleration using ML, Computer Vision, Digital Signal & Image Processing, Hardware-Software Co-Design

Bhoj Reddy Engineering College for Women, Hyderabad, Telangana, India. June 2016 – September 2020

Bachelor of Technology | Electronics and Communication Engineering

Relevant Coursework: VLSI Design, Microprocessors and Microcontrollers, Digital Logic Design, Semiconductor Devices & Electronic Circuits

CERTIFICATIONS

- Fundamentals of System Verilog, System Verilog Projects, UVM Fundamentals from Namaste FPGA
- Design Certifications in CDC, STA, I2C, SPI, UART, AMBA AXI, AMBA APB in Verilog